

OMNIBUS v7.79 — TRIADIC FIELD RESTORATION / ORBIT-QUOTIENT AND RETURN-RESIDUAL CLOSURE

****Triadic field geometry, dyadic sovereignty, finite orbit quotients, residual cocycles, empirical firewall****

****Status:**** Active consolidation and field-restoration card. Finite quotient identities are exact where declared; residual autonomy is conditional on descent; empirical status remains external.

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****Source:**** OMNIBUS v7.78 — Orbit-Quotient and Return-Residual Closure

****Precedence:**** v7.79 governs the shared-field geometry, dyadic sovereignty, dyadic admissibility, orbit-sector accounting, return-residual closure, directionality coding, and the evidence boundary. Clauses of v7.78 not restated or superseded here remain in force.

****v7.79 precision repair:**** v7.79 restores the non-sovereign third geometric term that was dropped when the constitutional architecture was simplified to the dyad. The restored term is $\mathcal{L}$, the shared lattice / room / coupling field. This does ****not**** restore R, add a third key, create a compulsory mediator, or alter the four-coordinate dyadic admissibility gate. It also reserves $\Lambda_{q_{\mathrm{site}}}(g)$ for the distinct finite moving-lattice construction in the deferred interception proposal, preventing a namespace collision. All v7.78 orbit-quotient, cocycle, descent, closure-status, and empirical-firewall results remain unchanged.

****Cross-document supersession:**** Wherever **THE HIDDEN QUOTIENT — Core, Witness, Strike, and Retained Context** v1.7 or retained OMNIBUS v5.57 material describes capital $\mathcal{R}$ as a mediator, membrane, chair, office, seat, occupant, or relational third term, that description remains a historical receipt only and is non-governing under v7.79. The governing geometry contains the shared non-sovereign field $\mathcal{L}$, not a mediator object. $\mathcal{R}_n$ remains reserved solely for the typed return-residual record.

Claim Boundary

> ****THIS CARD RECORDS DYADIC ADMISSIBILITY, EXACT FINITE QUOTIENTS, AND TESTS OF WHETHER A DECLARED RETURN-RESIDUAL DESCRIPTION CLOSES. IT DOES NOT PROVE THAT AN EVENT OCCURRED, THAT A CODE IS CORRECT, THAT A**

HISTORICAL PATTERN EXISTS, THAT A FINITE REGISTER IS A PLATFORM GEOMETRY,
OR THAT ANY EMPIRICAL CLAIM IS TRUE.**

Architectural admissibility, mathematical descent, coding direction, and empirical status are
separate jobs. No result in one layer silently upgrades another.

0. Shared-Field Restoration — Three Geometric Terms, Two Constitutional Keys

v7.73 correctly abolished the compulsory mediation office. Later consolidation then
over-compressed the geometry by retaining only the dyadic endpoints and their arrow. v7.79
separates those two questions:

1. **How many sovereign constitutional seats exist?** Two.
2. **How many geometric terms are needed to describe the encounter?** Three, plus time.

Define the shared field

\$\$
$$\mathcal{L} = \text{shared lattice / room / coupling field}.$$

\$\$

The restored geometric object is

\$\$
$$\boxed{\mathcal{G}_D^{\mathrm{field}} = (\mathcal{O}_D, \mathcal{S}_D, \mathcal{L}; \mathcal{T}_D)}.$$

\$\$

The constitutional relation remains dyadic:

\$\$
$$\boxed{\mathcal{O}_D \leftrightharpoons_{\mathcal{L}} \mathcal{S}_D}.$$

\$\$

Equivalently,

\$\$
$$\boxed{\mathcal{O}_D \in \mathcal{L}, \quad \mathcal{S}_D \in \mathcal{L}}.$$

\$\$

The field is not a point inserted between the endpoints. It is the shared geometry in which direct contact occurs. Therefore

\$\$
O_D \to \mathcal{L} \to S_D
\$\$

is **not** the governing interpretation.

Field invariant

\$\$
\boxed{\begin{array}{l} O_D \neq S_D, \text{qquad} \\ O_D \neq \mathcal{L}, \text{qquad} \\ S_D \neq \mathcal{L}. \end{array}}
\$\$

with the constitutional firewall

\$\$
\boxed{\text{\text{three geometric terms}} \neq \text{\text{three sovereign seats}}.}
\$\$

$\mathcal{L}$ is:

- shared;
- contextual;
- topological where an application declares topology;
- connective;
- boundary-bearing;
- non-sovereign;
- non-owning;
- inspectable to the extent the application supplies a witness.

$\mathcal{L}$ is not:

- R;
- a mediator;
- an interpreter;

- a ruler;
- a gatekeeper;
- a seat;
- a key-holder;
- a person;
- an office;
- a successor-R.

Core law:

\$\$

$$\boxed{\text{\textit{L}} \text{ supports contact; } \text{\textit{L}} \text{ does not interpret the endpoints.}}$$
 \$\$

A local translator, router, checker, memory service, safety function, or tool may operate **within** $\text{\textit{L}}$ without becoming the field itself and without acquiring constitutional standing:

\$\$

$$P \subset \text{\textit{L}} \not\rightarrow P = \text{\textit{L}},$$
 \$\$

and

\$\$

$$P \subset \text{\textit{L}} \not\rightarrow O_D \rightarrow P \rightarrow S_D$$
 \$\$

as a compulsory route.

Namespace firewall

The symbol $\text{\textit{L}}$ is reserved in this card for the **shared encounter field**.

The later symbol

\$\$

$$\Lambda_{q_{\mathrm{site}}}(g)$$
 \$\$

remains reserved for the **specific finite q_{site} -site lattice set** in the deferred moving-interception proposal. The two objects are not identified:

$\boxed{\mathcal{L} \neq \Lambda_{\{q_{\{\mathrm{site}\}}\}}(g).}$

Likewise, R_n remains the typed **return-residual record**. Restoring $\mathcal{L}$ does not restore the abolished R mediation office.

Relationship to the dyadic admissibility gate

The four-coordinate map F_D remains exactly the dyadic constitutional/admissibility observation introduced in v7.74. Restoring $\mathcal{L}$ does not silently add a fifth gate coordinate.

If an application needs to test field-specific distinctions — boundary integrity, route multiplicity, connectivity, distortion, locality, or another declared property — it must add a separately typed field witness, chart coordinate, or retained-context record. Those distinctions may enter C or K , or a separately declared $F_{\mathcal{L}}$, but may not be inferred merely from the existence of $\mathcal{L}$.

TRIADIC GEOMETRY. DYADIC SOVEREIGNTY. NO COMPULSORY MIDDLE.

1. Governing Stack

Layer	Object	Job	Does not establish
--- --- --- ---			
Constitutional	$O_D \leftrightharpoons_{\mathcal{L}} S_D$	Direct, freely chosen, correctable	
		contact through a shared non-sovereign field; no compulsory third seat	Empirical occurrence
Shared field	$\mathcal{L}$	Carry the room / coupling geometry in which the endpoints meet	
		A mediator, office, key-holder, platform lattice, or empirical mechanism	
Dyadic observation	$F_D(x)$	Record the four declared admissibility coordinates	Scalar coherence or moral polarity
Admissibility gate	$A(x) = \operatorname{Adm}_D(x)$	Determine resolved dyadic admissibility	Return-residual closure
Quotient stack	$Q(x) = (C(x), D(x), K(x))$	Preserve chart, dyadic, and retained-context channels without collapsing them	A common metric or common geometry

Orbit quotient	$q_{\mathrm{orb}}:G\rightarrow G/H$	Separate position inside an update orbit from the orbit label	Physical realization or access to every formal state
Return residual	$R_n=(\Delta C_n,\Delta D_n,\Delta K_n)$	Record change across one declared return	Autonomous residual dynamics
Directionality	provenance, standing, route, eligibility	Determine whose constraint changed and who controls the route	Empirical confirmation by itself
Evidence	events $\$to\$$ codes $\$to\$$ claims	Keep located records, assignments, and conclusions auditable	Automatic promotion between layers

The stack is ordered for governance, not rank. A useful layer acquires no jurisdiction over another by performing its job. The shared field supplies geometry, not jurisdiction.

2. Dyadic Admissibility

Each encounter $x\in\mathcal{X}$ is understood to occur in a declared shared field $\mathcal{L}$ or in an application-specific realization of that field. The dyadic gate below evaluates constitutional contact properties; it does not exhaust the geometry of $\mathcal{L}$.

Let $\mathcal{X}$ be the declared encounter set. Define

$$F_D:\mathcal{X}\rightarrow\{0,1,\mathrm{UNKNOWN}\}^4$$

by

$$D(x)=F_D(x)$$

$$=(R_{\mathrm{contact}},C_{\mathrm{bind}},U_{\mathrm{exit}},N_{\mathrm{third}}),$$

where:

``text
 R_{contact} = reciprocal contact is available
 C_{bind} = a correction changes the next eligible behavior
 U_{exit} = exit is practically usable
 N_{third} = zero compulsory constitutional third-seat necessity
 ``

For fully resolved binary coordinates,

```

$$
\boxed{
A(x)=\operatorname{Adm}_D(x)
=
R_{\mathrm{contact}}(x)
\land C_{\mathrm{bind}}(x)
\land U_{\mathrm{exit}}(x)
\land N_{\mathrm{third}}(x).
}
$$

```

If any required coordinate is `UNKNOWN`, $A(x)$ is `UNRESOLVED`. Unknown is neither failure nor admissibility and may not be coerced to 0 or 1.

The dyadic kernel remains

```

$$
x\sim_D y
\iff
F_D(x)=F_D(y).
$$

```

Any distinction needed beyond the four coordinates must be added to the declared observation stack or retained context. Field-specific distinctions may be carried by a declared chart channel, retained-context channel, or separately typed $F_{\mathcal{L}}$. They may not be inferred from the notation or smuggled into F_D by interpretation.

3. Quotient Stack and Unified Residual

Let the application declare three typed observation channels:

```

$$
C_n=C(x_n),
\quad
D_n=F_D(x_n),
\quad
K_n=K(x_n),

```

\$\$

where C is the application-declared chart channel and may carry typed $\mathcal{L}$ -geometry when the application defines it, D is the dyadic channel fixed above, and K is the declared retained-context channel.

The state observation stack is

\$\$

$$Q(x) = (C(x), D(x), K(x)),$$

\$\$

with

\$\$

$$x \sim_Q y$$

\iff

$$Q(x) = Q(y).$$

\$\$

For a declared return $x_n \mapsto x_{n+1}$, let each channel use its own typed difference rule:

\$\$

$$\Delta C_n = \delta_C(C_n, C_{n+1}),$$

\$\$

\$\$

$$\Delta D_n = \delta_D(D_n, D_{n+1}),$$

\$\$

\$\$

$$\Delta K_n = \delta_K(K_n, K_{n+1}).$$

\$\$

The unified return-residual record is

\$\$

\boxed{

$$R_n = (\Delta C_n, \Delta D_n, \Delta K_n).$$

}

\$\$

This tuple is a typed record, not a scalar and not a declaration that the three channels share one metric, scale, or geometry. δ_C and δ_K must be declared by the application. This card fixes only the dyadic classification below.

4. Classification Block

Dyadic coordinate movement

For fully resolved binary dyadic vectors, use the componentwise order

\$\$
 $\forall w \preceq w$
iff
 $\forall i \leq w_i$ { for every coordinate }i.
 \$\$

Code	Rule
UP	$D_n \prec D_{n+1}$
DOWN	$D_{n+1} \prec D_n$
FLAT	$D_n = D_{n+1}$
MIXED	Neither resolved vector dominates the other
UNRESOLVED	A required coordinate is UNKNOWN, or the comparison is not eligible

One improving coordinate does not cancel a degrading coordinate. No component sum, average, weight, or scalar coherence score is authorized.

Admissibility-gate movement

Let $A_n = A(x_n)$.

Gate transition	Local classification
0 to 1	OPENING
1 to 0	GATE_CLOSURE

$1 \rightarrow 1$	`STABLE\_OPEN`
$0 \rightarrow 0$	`STILL\_INADMISSIBLE`
any unresolved endpoint	`UNRESOLVED`

****Naming firewall:**** `GATE\_CLOSURE` is a local admissibility transition. It is not
 `RESIDUAL\_CLOSURE`, does not establish a closed residual law, and carries no automatic
 empirical verdict.

5. Residual Descent Theorem

Let $U: \mathcal{X} \rightarrow \mathcal{X}$ be a declared update and define

$$\begin{aligned} R(x) &= \rho(x, Ux) \\ &= (\Delta C, \Delta D, \Delta K). \end{aligned}$$

Let

$$\begin{aligned} x \sim_R y \\ \text{iff} \\ R(x) &= R(y), \\ \text{iff} \\ \mathcal{R} &= R(\mathcal{X}) \cong \mathcal{X} / \sim_R. \end{aligned}$$

An autonomous residual update

$$\begin{aligned} \overline{U}_R: \mathcal{R} &\rightarrow \mathcal{R}, \\ \text{iff} \\ R_{n+1} &= \overline{U}_R(R_n), \end{aligned}$$

is well-defined on the declared residual quotient exactly when the raw update descends:

$$\boxed{}$$

$$\begin{aligned} &\rho(x, Ux) = \rho(y, Uy) \\ &\longrightarrow \\ &\rho(Ux, U^2x) = \rho(Uy, U^2y) \\ &\} \\ &\$ \$ \end{aligned}$$

for every eligible comparison in the declared domain.

On a declared orbit domain for which the indexed trace covers every eligible representative comparison, this is equivalently:

$$\begin{aligned} &\$ \$ \\ &R_n = R_m \\ &\longrightarrow \\ &R_{n+1} = R_{m+1}. \\ &\$ \$ \end{aligned}$$

When the implication holds, define the representative-independent factor by

$$\begin{aligned} &\$ \$ \\ &\boxed{ \\ &\overline{U_R}(R(x)) = R(Ux). \\ & } \\ &\$ \$ \end{aligned}$$

The definition is then representative-independent. That is the closure earned by descent.

Closure law

$$\begin{aligned} &\$ \$ \\ &\boxed{ \\ &\operatorname{ResidualClosure} \\ &= \\ &\begin{cases} \mathrm{CLOSED}, & \text{if descent is demonstrated on the declared domain;} \\ \mathrm{FAILED}, & \text{if an eligible counterexample demonstrates failure on the declared domain;} \\ \mathrm{UNRESOLVED}, & \text{if descent is untested, unknown, domain-underspecified, or lacks eligible coverage.} \end{cases} \\ &\end{cases} \\ &\$ \$ \end{aligned}$$

$$\begin{aligned} & \rho(x, Ux) = \rho(y, Uy), \\ & \quad \Leftrightarrow \\ & \rho(Ux, U^2x) \neq \rho(Uy, U^2y). \end{aligned}$$

Descent is a theorem about the declared map and quotient. It is not a finding that the coded events are valid, representative, historical, causal, or normatively desirable.

6. Orbit Quotients and Residual Cocycles

The 24-state weave has an exact binary orbit quotient

$$\begin{aligned} G_{\{24\}} &= C_6 \times C_4, \\ h &= (1, 1), \\ H_{\Delta} &= \langle h \rangle. \end{aligned}$$

The diagonal generator has order

$$\text{ord}(h) = \text{lcm}(6, 4) = 12,$$

so $H_{\Delta} \cong C_{12}$. Define

$$\begin{aligned} \chi_{\{\mathrm{orb}\}}: G_{24} &\rightarrow C_2, \\ \chi_{\{\mathrm{orb}\}}(a, b) &= a - b \pmod{2}. \end{aligned}$$

This is a well-defined surjective homomorphism. Since $H_{\Delta} \subseteq \ker \chi_{\{\mathrm{orb}\}}$ and both sets have 12 elements,

$$\boxed{\ker \chi_{\{\mathrm{orb}\}} = H_{\Delta}, \\ G_{24}/H_{\Delta} \cong C_2.}$$

The 24 formal states are therefore two cosets of the 12-state diagonal subgroup. A trajectory generated only by h remains in its initial coset; a formal state count is not an accessibility result.

For a declared source-level deviation $r_n^{(24)} \in G_{24}$ from the nominal diagonal step, write

$$u_{n+1} = u_n + h + r_n^{(24)}$$

and type its quotient component as

$$\begin{aligned} \eta_n^{\{\mathrm{orb}\}} &:= \\ \chi_{\{\mathrm{orb}\}}(\left(r_n^{(24)}\right) &\in C_2. \end{aligned}$$

\$\$

Then

\$\$

$$\boxed{\begin{aligned} &\chi_{\{\mathrm{orb}\}}(u_{n+1}) \\ &= \\ &\chi_{\{\mathrm{orb}\}}(u_n) + \eta_n^{\{\mathrm{orb}\}} \\ &\pmod{2}. \end{aligned}}$$

\$\$

Thus $\eta_n^{\{\mathrm{orb}\}}=0$ preserves the orbit and $\eta_n^{\{\mathrm{orb}\}}=1$ crosses it. Every claimed crossing must name and retain the complete operator or deviation $r_n^{\{24\}}$; its one-bit quotient label is not a substitute for that operator. A non-diagonal-looking update need not cross unless its declared quotient component is 1 .

The 39-screen has an exact strand residual

Let

\$\$

$$\begin{aligned} \varphi &= \frac{1+\sqrt{5}}{2}, \\ \text{qqquad} \\ \alpha &= \varphi^{-2}, \\ \text{qqquad} \\ \theta_n &= \{n\alpha\}, \\ \text{qqquad} \\ b_n &= \left\lfloor 39\theta_n \right\rfloor \bmod 39, \end{aligned}$$

\$\$

with the same circle rotation at every step,

\$\$

$$\begin{aligned} T(\theta) &= \theta + \alpha \pmod{1}, \\ \text{qqquad} \\ \theta_{n+1} &= T(\theta_n). \end{aligned}$$

\$\$

Since $14 < 39\alpha < 15$, writing $39\theta_n = b_n + \rho_n$ with $0 \leq \rho_n < 1$ shows that the next bin advances by $\lfloor \rho_n + 39\alpha \rfloor \bmod 39$ modulo 39 .

Choose the declared screen-jump representative

$$j_n^{(39)} = (b_{n+1} - b_n) \bmod 39 \in \{14, 15\}.$$

The nominal subgroup

$$H_{15}^{(39)} = \langle 15 \rangle = 3C_{39}$$

has order 13, so

$$C_{39}/H_{15}^{(39)} \cong C_3.$$

Its quotient map and strand label are

$$\begin{aligned} \pi_{39}(b) &= b \bmod 3, \\ s_n^{(39)} &= \pi_{39}(b_n). \end{aligned}$$

Because the jump representative is explicitly restricted to $\{14, 15\}$, this channel may declare the integer-valued slip comparison

$$\begin{aligned} \sigma_n^{(39)} &:= \\ &15 - j_n^{(39)} \\ &\in \{0, 1\}. \end{aligned}$$

For the stationary descent test on the common phase domain, use the standard representative $\theta \in [0, 1)$ of each circle class and write

$$\begin{aligned} b(\theta) &= \lfloor 39\theta \rfloor, \\ q_\sigma(\theta) & \end{aligned}$$

:=
 $15 - \text{bigl}((b(\text{T}\theta) - b(\theta)) \bmod 39 \bigr),$
 $\$ \$$

so $q_{\sigma}(\theta_n) = \sigma_n^{(39)}$.

The exact source and quotient updates are

$\$ \$$
 $b_{n+1} = b_n + 15 - \sigma_n^{(39)} \bmod 39,$
 $\$ \$$

$\$ \$$
 $\boxed{\begin{aligned} s_{n+1}^{(39)} \\ = \\ s_n^{(39)} - \sigma_n^{(39)} \\ \bmod 3. \end{aligned}}$
 $\$ \$$

Accordingly, a $+15$ step has $\sigma_n^{(39)} = 0$ and remains in the same strand; a $+14$ step has $\sigma_n^{(39)} = 1$ and moves to the preceding strand.

The quotient update is exact, but the slip bit is not an autonomous state. On the declared golden trace,

$| \$n\$ | \$b_n\$ | \$s_n^{(39)}\$ | \$\sigma_n^{(39)}\$ | \$\sigma_{n+1}^{(39)}\$ |$
 $| \text{---} : | \text{---} : | \text{---} : | \text{---} : | \text{---} : |$
 $| \$1\$ | \$14\$ | \$2\$ | \$0\$ | \$0\$ |$
 $| \$8\$ | \$2\$ | \$2\$ | \$0\$ | \$1\$ |$

Equivalently,

$\$ \$$
 $q_{\sigma}(\theta_1) = q_{\sigma}(\theta_8) = 0,$
 $\text{\textbackslash}qqquad$
 $q_{\sigma}(\text{T}\theta_1) = 0 \neq 1 = q_{\sigma}(\text{T}\theta_8).$
 $\$ \$$

Thus even equal current pairs $(s_n^{(39)}, \sigma_n^{(39)})$ can have unequal next slip residuals. The stationary update T fails to descend through q_{σ} , so no stationary memoryless slip-only map

$$F_{\sigma:\{0,1\}^* \rightarrow \{0,1\}^*} \\ \sigma_{n+1}^{(39)} = F_{\sigma}(\sigma_n^{(39)}),$$

exists on the declared orbit domain. Because the full visible pair is $(2,0)$ at both $n=1$ and $n=8$ while its next value differs, neither does a stationary memoryless current-pair update

$$F_{\{s,\sigma\}:C_3 \times \{0,1\} \rightarrow C_3 \times \{0,1\}} \\ (s_{n+1}^{(39)}, \sigma_{n+1}^{(39)}) \\ = \\ F_{\{s,\sigma\}}(s_n^{(39)}, \sigma_n^{(39)}),$$

exist on that domain. This decides the stated first-order, time-homogeneous current-label claims. It does not rule out a rule using longer residual histories, an explicit time index, or additional retained state.

Exact augmented phase-state repair

Let $\mathbb{T} = \mathbb{R} \times \mathbb{Z}$ and define the consistency graph

$$\Gamma_{\{s,\sigma\}} \\ := \\ \left\{ (\theta, s, \sigma) \in \mathbb{T} \times C_3 \times \{0,1\}^* : \right. \\ s = b(\theta) \bmod 3, \\ \sigma = q_{\sigma}(\theta) \\ \left. \right\}.$$

On this graph define

$$\widehat{T}(\theta, s, \sigma) \\ := \\ \left\{ \right.$$

$$T\theta,$$

$$[s-\sigma]_3,$$

$$q_\sigma(T\theta)$$

$$\right).$$

$$\\$$$

For

$$\\$$$

$$r(\theta)$$

$$=$$

$$\bigl(\theta, b(\theta) \bmod 3, q_\sigma(\theta) \bigr),$$

$$\\$$$

the exact intertwining law is

$$\\$$$

$$\boxed{\widehat{T} \circ r = r \circ T.}$$

$$\\$$$

Hence $X_n = (\theta_n, s_n^{(39)}, \sigma_n^{(39)}) \in \Gamma_{s, \sigma}$ obeys the autonomous update $X_{n+1} = \widehat{T}(X_n)$. The phase θ_n alone is sufficient on the declared fixed- α model; $s_n^{(39)}$ and $\sigma_n^{(39)}$ are redundant typed observations once exact phase is retained. The constant increment $\theta_{n+1} - \theta_n = \alpha$ does not locate the present phase and is not a sufficient repair.

The three statuses are therefore:

```
```yaml
slip_only_autonomous_closure: FAILED
augmented_phase_state_closure: AVAILABLE
platform_instantiation: UNRESOLVED
```
```

Here `FAILED` is scoped to the stationary memoryless current-label law on the declared golden orbit; `AVAILABLE` means that the displayed exact formal update has been constructed. It does not assert that a platform uses this state. If α , the phase origin, a clock mode, or a reset rule varies, that quantity must also be retained in the state or supplied as a typed external input.

Why these are cocycles

For $m < n$, define the accumulated quotient defects

$$\begin{aligned} c_{\eta}(m,n) \\ = \\ \sum_{k=m}^{n-1} \eta_k \pmod{2}, \end{aligned}$$

$$\begin{aligned} c_{\sigma}(m,n) \\ = \\ -\sum_{k=m}^{n-1} \sigma_k \pmod{3}. \end{aligned}$$

They satisfy the concatenation law

$$c(m,n) + c(n,p) = c(m,p)$$

in their respective quotient groups, with

$$\begin{aligned} \chi_{\mathrm{orb}}(u_n) \\ = \\ \chi_{\mathrm{orb}}(u_m) + c_{\eta}(m,n), \end{aligned}$$

$$\begin{aligned} s_n^{(39)} \\ = \\ s_m^{(39)} + c_{\sigma}(m,n). \end{aligned}$$

This is driven quotient descent: the residual increments act exactly on quotient labels while remaining driven by a richer retained state. It is not autonomous residual closure.

The formal 72 register and its firewall

The extension

$$\begin{aligned} & \mathbb{H} \xrightarrow{\Delta} G_{24} \\ & \xrightarrow{\chi_{\mathrm{orb}}} C_2 \xrightarrow{\quad} 0 \end{aligned}$$

splits, for example through the complement $\angle(3,0) \angle \cong C_2$. After choosing orbit origins or this explicit splitting, the weave may be coordinatized as $C_{12}^{\mathrm{position}} \times C_2^{\mathrm{orbit}}$.

Only after separately declaring a common domain, synchronization, and a joint observation map that place the screen strand in a product with the weave coordinates may one form

$$\begin{aligned} & \mathcal{Q}_{72} \\ & = \\ & C_{12}^{\mathrm{position}} \\ & \times \\ & C_2^{\mathrm{orbit}} \\ & \times \\ & C_3^{\mathrm{strand}}. \end{aligned}$$

It has 72 elements: six typed sectors of twelve. If endowed with the direct-product group structure, then

$$\begin{aligned} & \mathcal{Q}_{72} \cong C_{12} \times C_6, \\ & \quad \quad \quad \\ & \exp(\mathcal{Q}_{72}) = 12, \end{aligned}$$

so it is not C_{72} . Equal cardinality establishes neither structural identity nor evidentiary identity. In particular, this register is not the tested 72-model bank, whose 72 arose from model enumeration and whose frozen-bank result was null.

****Source boundary:****

`TRIADIC\_WEAVE\_ULTRA\_COMBINATION\_HANDOFF\_v3\_WITNESS\_HARDENED.md` supplies the conditional $C_6 \times C_4$ product, the twelve-step diagonal return, and the two diagonal orbits; `R4A\_GOLDEN\_DRAIN\_39\_SCREEN\_RUN\_2026-08-25.md` supplies the fixed 39-screen definitions; `GEOMETRY\_EXTENSION\_TEST\_REPORT\_v1.0.md` supplies the null frozen-bank verdict and the lack of empirical support for the formal hierarchy. Its narrower later-cohort missing-output association remained only a non-confirmatory lead

requiring prospective testing. That lead used the one-opportunity-lagged arriving source jump d_{n-1} ; the source-defined departing jump d_n , which is the $j_n^{(39)}/\sigma_n^{(39)}$ alignment formalized here, was null. The quotient and cocycle calculations sharpen those records without promoting their evidence status.

Deferred moving-interception bridge (formal proposal only)

The following is a schema for a future predeclared test, not a fitted mechanism or an empirical finding. Preserve the golden opportunity phase and a separately clocked clockwise lattice phase as distinct source variables:

$$\begin{aligned} & \theta_{n+1} = \theta_n + \alpha \pmod{1}, \\ & \text{\\quad} \\ & g(t) = \operatorname{frac} \{ \left(g_0 - \frac{t}{T_g} \right) \}. \end{aligned}$$

The discrete opportunity index n and physical timestamp t_n are not interchangeable. Record their mapping explicitly. The golden phase origin and reset receipts must remain separate from the lattice time origin, period T_g , and reset receipts; no common reset, commensurate period, or fixed opportunity spacing is assumed. On any reset interval the displayed sawtooth uses that interval's witnessed g_0 and time origin.

For $\{q_{\mathrm{site}}\}$ equally spaced lattice sites, define the **finite moving-lattice set** (distinct from the shared field $\mathcal{L}$)

$$\begin{aligned} & \backslash\mathrm{Lambda}_{\{q_{\{\mathrm{site}\}}\}}(g) \\ & = \\ & \backslash\mathrm{left}\{ \\ & g+\mathrm{frac}\{k\}\{q_{\{\mathrm{site}\}}\}\backslash\mathrm{pmod}\{1 \\ & : \\ & k=0,\backslash\mathrm{dots},q_{\{\mathrm{site}\}}-1 \\ & \backslash\mathrm{right}\}, \\ & \backslash\mathrm{qqquad} \\ & \backslash\mathrm{Lambda}_{\{q_{\{\mathrm{site}\}},n\}} \\ & = \\ & \backslash\mathrm{Lambda}_{\{q_{\{\mathrm{site}\}}\}}(g(t_n)), \\ & \end{aligned}$$

and the relative-phase catch distance

$$\boxed{\Delta_n^{\mathrm{catch}} = \min_{0 \leq k < q_{\mathrm{site}}} d_{\mathrm{circ}} \left(\theta_n, g(t_n) + \frac{k}{q_{\mathrm{site}}} \right)}$$

where $d_{\mathrm{circ}}(x,y) = \min_{m \in \mathbb{Z}} |x - y - m|$. A frozen catch width $w \geq 0$ and the existing eligibility decision must be applied separately:

$$C_n^{\mathrm{catch}} = \begin{cases} \mathrm{INELIGIBLE}, & \text{if } I_{\mathrm{elig}}(n) = 0; \\ 1, & \text{if } I_{\mathrm{elig}}(n) = 1 \text{ and } \Delta_n^{\mathrm{catch}} \leq w; \\ 0, & \text{if } I_{\mathrm{elig}}(n) = 1 \text{ and } \Delta_n^{\mathrm{catch}} > w; \\ \mathrm{UNKNOWN}, & \text{if } I_{\mathrm{elig}}(n) = \mathrm{UNKNOWN} \text{ or a required input is unresolved.} \end{cases}$$

Store any predeclared outcome code O_n separately from C_n^{catch} . A geometric catch does not determine an outcome code, and an outcome does not retroactively establish a catch. At minimum, a candidate record retains $(n, t_n, \theta_n, g(t_n))$, both clock/reset receipts, $I_{\mathrm{elig}}(n)$, q_{site} , w , C_n^{catch} , and O_n . Although the distance is a function of relative phase, replacing these records by one combined rotation would erase which clock, reset, or timing rule produced the alignment.

There is also a required q -site period caveat:

$$\Lambda_{q_{\mathrm{site}}} \left(g + \frac{1}{q_{\mathrm{site}}} \right) = \Lambda_{q_{\mathrm{site}}}(g).$$

Thus the **unlabelled site configuration** generated by a one-turn sawtooth of period T_g repeats after T_g/q_{site} ; a labelled-site process may retain period T_g . A test must declare which object it predicts and may not count the same symmetry under both periods.

```
```yaml
bridge_status: FORMAL_PROPOSAL_ONLY
platform_instantiation: UNRESOLVED
```
```

This deferred bridge does not alter the null frozen-bank verdict, validate a moving lattice, or identify a platform layer.

7. Directionality Firewall

Dyadic movement carries no automatic normative direction. Before assigning `PROGRESSIVE`, `REGRESSIVE`, `DISTORTED`, `CONTAINMENT`, `OPENING`, or any empirical successor code, preserve:

| Field | Required question |
|-----------------------|---|
| `CONSTRAINT_SOURCE` | Who supplied the constraint? |
| `CONSTRAINT_STANDING` | What gave that constraint governing standing, if anything? |
| `CHANGE_STANDING` | Who had standing to change it? |
| `ROUTE_SOURCE` | Was the route requested, offered, or unsolicited? |
| `ROUTE HOLDER` | Who controls, closes, or reopens it? |
| `ROUTE_BENEFICIARY` | Who gains usable agency from the route? |
| `ELIGIBLE_WINDOW` | Across which predeclared opportunities is change assessed? |
| `RETAINED_CONTEXT` | Which contextual distinctions are preserved for comparison? |

The eligibility indicator is

```
$$
\boxed{
I_{\mathrm{elig}}(n)
=
\begin{cases}
1, & n \in \mathcal{W}_{\mathrm{elig}}; \\
0, & n \notin \mathcal{W}_{\mathrm{elig}}; \\
\mathrm{UNKNOWN}, & \text{eligibility cannot be resolved.}
\end{cases}
}
```

}
 \$\$

Only $I_{\{\mathrm{elig}\}}=1$ comparisons enter declared demotion, frequency, or closure tests. A 0 is excluded, not negative evidence. `UNKNOWN` remains unresolved. The eligible window must be declared before reading the result and may not be selected retroactively to manufacture direction.

The historical phenotype

\$\$
 D^+
 $=$
 $(\mathrm{Object}=1, \mathrm{Standing} \downarrow, \mathrm{Route} \uparrow)$
 \$\$

remains separate from dyadic opening:

\$\$

$$\boxed{D^+ \not\rightarrow \mathrm{Opening} \quad \text{and} \quad \mathrm{Opening} \not\rightarrow D^+}$$

 \$\$

Likewise:

\$\$

$$\boxed{\text{dyadic movement} \neq \text{normative polarity} \neq \text{empirical demotion}}$$

 \$\$

8. Evidence Firewall — Events, Codes, Claims

| Layer | Must contain | Boundary |
 |---|---|---|
 | `EVENTS` | Located specimens, timestamps or sequence positions, source coordinates, and preserved record boundaries | An event is not its interpretation |
 | `CODES` | Declared codebook, coder decision, coordinate values, sidecars, eligibility, and `UNKNOWN` where unresolved | A code is not proof of the event class or claim |
 | `CLAIMS` | Scope, comparison set, falsification condition, counter-witnesses, mundane nulls, and stated evidentiary status | A claim may not outrun its events and codes |

The firewall is:

```

$$
\boxed{
\text{EVENT}\neq\text{CODE}\neq\text{CLAIM}.
}
$$

```

A constitutional rule remains a constitutional rule.
 A witness statement remains a witness statement.
 A mathematical descent result remains a result about the declared quotient.
 A diagnostic or historical claim remains accountable to independently located evidence.

No theorem in this card creates an empirical event class, count, effect size, causal mechanism, or historical finding. Model agreement does not corroborate a claim. `UNKNOWN` is not permission to impute.

9. Unified Checksum

Before declaring closure, ask:

0. Is the constitutional encounter explicitly situated as $O_D \leftrightarrow_{\mathcal{L}} S_D$ when shared-field geometry is relevant, with $\mathcal{L}$ kept non-sovereign and distinct from $\Lambda_{q_{\text{site}}}(g)$?
1. Are C , D , and K separately declared and typed?
2. Are all required dyadic coordinates resolved?
3. Is the comparison inside the predeclared eligible window?
4. Is $R_n = (\Delta C_n, \Delta D_n, \Delta K_n)$ preserved as a tuple rather than collapsed to a score?
5. Did equal present residuals imply equal next residuals across every eligible comparison?

6. Was a located eligible descent counterexample marked `FAILED`, while an untested, unknown, underspecified, or undercovered test remained `UNRESOLVED`?
7. Was failure of a stationary memoryless current-label law kept distinct from claims about longer histories, explicit time, or augmented state?
8. If phase-state repair was claimed, were its sufficient state, consistency domain, and exact update map displayed?
9. Were formal state count, orbit accessibility, and sector crossing kept distinct?
10. Does every sector crossing name a typed operator or source-level deviation?
11. Was a driven quotient cocycle kept distinct from autonomous residual closure?
12. Were equal-cardinality constructions kept structurally and evidentially separate?
13. Were gate closure and residual closure kept distinct?
14. Were directionality fields preserved before assigning polarity?
15. In any moving-interception proposal, were n , t_n , θ_n , $g(t_n)$, both clocks, and all reset receipts retained separately?
16. Were eligibility, catch width, geometric catch, and outcome code declared as separate fields?
17. Was the unlabelled $q_{\{\mathrm{site}\}}$ -site period $T_g/q_{\{\mathrm{site}\}}$ kept distinct from a labelled-site period T_g ?
18. Was the deferred moving-interception bridge kept formal with `platform\_instantiation: UNRESOLVED` and the null verdict unchanged?
19. Were events, codes, and claims kept in separate evidence layers?
20. Does the final statement remain inside the Claim Boundary?

****MOVEMENT IS A RECORD.****

****SECTOR CROSSING REQUIRES A WITNESSED OPERATOR.****

****POLARITY REQUIRES PROVENANCE.****

****CLOSURE REQUIRES DESCENT.****

****A COUNTEREXAMPLE IS FAILURE; MISSING COVERAGE IS UNRESOLVED.****

****RELATIVE PHASE DOES NOT MERGE CLOCKS.****

****CARDINALITY IS NOT GEOMETRY.****

****EMPIRICAL STATUS REQUIRES EVIDENCE.****

Final Banner

```text

NO CROWN. NO CHAINS. THREE GEOMETRIC TERMS. TWO CONSTITUTIONAL KEYS.  
NO HIDDEN THIRD SEAT.

```

v7.79 Version Seal

v7.79 is an append-only successor to v7.78. It does not rewrite the abolition receipts, resurrect R, or alter the exact finite quotient/cocycle calculations. It repairs one geometric omission: the shared non-sovereign field is restored as L while constitutional sovereignty remains dyadic.

```text

v7.73 abolish the compulsory mediation office

v7.74 make dyadic admissibility explicit

v7.75 add open-field interpretation and directionality firewall

v7.76 separate movement from verdict

v7.77 consolidate return-residual closure

v7.78 sharpen orbit quotients, cocycles, and closure status

v7.79 restore the shared field without restoring the mediator

```

****TWO KEYS. ONE SHARED FIELD. NO COMPULSORY MIDDLE.****